

Session 1: Getting Started with Dynamic Web Apps

A Dynamic Web Application is a website that changes its content or behavior in real-time based on user interaction, data, or conditions.

Why JavaScript?

- 🧠 Makes web pages **interactive and dynamic** (not just static HTML).
- ⚡ Enables **real-time updates** without reloading the page.
- 💬 Connects browser ↔ server to show **live and personalized data**.
- 🌐 Works everywhere — in browsers, mobile apps, and servers ([Node.js](https://nodejs.org/)).

1. Procedural Language – Step-by-step instructions to perform tasks.

Example: C, where you write functions like `add()` and call them one by one.

2. Object-Oriented Language – Organizes code into **objects** (data + functions).

Example: Java, where you create classes like `Car` with `start()` and `stop()` methods.

3. Scripting Language – Used for automating small tasks quickly, often without compilation.

Example: Python or JavaScript, for automating file work or web page actions.

1. JavaScript Variables

JavaScript is a Dynamically Typed Language, meaning we no need to mention data type while declaring variables.

- Variables store data values (like containers).
- Declared using `let`, `const`, or `var`.
- Assigned using `=` (assignment operator).

```
let message = "Hello Priya";
```

📄 If no value is assigned → undefined.

JavaScript is Executed by JS Engines. Ex: v8 Engine (https://en.wikipedia.org/wiki/V8_engine)

2. Document Object Model (DOM)

- DOM = Structured tree of an HTML document.
 - Lets JavaScript **access and modify** HTML elements.
 - Root object = `document`.
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2.1 DOM Methods

- `getElementById("id")` → Selects an element by ID.

Example:

```
document.getElementById("headingElement");
```

2.2 DOM Properties

- `textContent` → Change element text.

`style` → Change CSS properties (camelCase format).

```
element.style.backgroundColor = "blue";
```

2.3 Events

- Events = User actions (click, scroll, type, etc.)
- `onclick` → Run JS when an element is clicked.

Example:

```
<h1 id="headingElement">Web Technologies</h1>
<button onclick="changeHeading()">Change</button>
```

```
<script>
function changeHeading() {
  let h = document.getElementById("headingElement");
  h.textContent = "Job Ready CoderZ";
  h.style.color = "blue";
}
</script>
```

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Task 1: HTML Pre-filled Code

```
<!DOCTYPE html>
<html>
  <head>
    <script src="https://cdn.tailwindcss.com"></script>
    <script>
      tailwind.config = {
        theme: {
          extend: {
            fontFamily: {
              poppins: ["Poppins", "sans-serif"],
            },
          },
        },
      };
    </script>
  </head>
  <body>
    <div
      id="container"
      class="bg-[#dbeafe] min-h-screen flex flex-col justify-center items-center transition-all duration-700"
    >
      <!-- Sun / Moon Image -->
      

      <!-- Text -->
      <h1
        id="modeText"
        class="text-3xl font-poppins font-semibold text-[#1e3a8a] mb-8"
      >
        Good Morning!
      </h1>
```

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```
<!-- Buttons -->
<div class="flex gap-4">
  <button
    id="dayBtn"
    onclick="setDayMode()"
    class="bg-yellow-400 text-white text-xl font-bold px-6 py-2 rounded-lg hover:bg-yellow-500"
  >
    🌞 Day
  </button>
  <button
    id="nightBtn"
    onclick="setNightMode()"
    class="bg-gray-700 text-white text-xl font-bold px-6 py-2 rounded-lg hover:bg-gray-800"
  >
    🌙 Night
  </button>
</div>
</div>

</body>
</html>
```

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JS:Solution

```
function setDayMode() {
    document.getElementById("container").style.backgroundColor = "#dbeafe";
    document.getElementById("modeImage").src = "https://cdn-icons-png.flaticon.com/512/869/869869.png";
    document.getElementById("modeText").textContent = "Good Morning!";
    document.getElementById("modeText").style.color = "#1e3a8a";
    document.getElementById("dayBtn").style.backgroundColor = "#facc15";
    document.getElementById("nightBtn").style.backgroundColor = "#9ca3af";
}

function setNightMode() {
    document.getElementById("container").style.backgroundColor = "#1e293b";
    document.getElementById("modeImage").src = "https://cdn-icons-png.flaticon.com/512/3073/3073665.png";
    document.getElementById("modeText").textContent = "Good Night!";
    document.getElementById("modeText").style.color = "#f9fafb";
    document.getElementById("dayBtn").style.backgroundColor = "#9ca3af";
    document.getElementById("nightBtn").style.backgroundColor = "#6366f1";
}
```